SHABANA A

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SUMMARY

B.Tech Graduate in Information Technology with a minor in ECE. AI Researcher specializing in Computer Vision, Deep Learning and Open-Vocabulary Object Detection (OVOD). Experienced in developing end-to-end Machine Learning workflows from customized CPU-optimized training to cloud-deployed Streamlit applications. Published original benchmarks on YOLO26, YOLOv11, and YOLO-World models with DOI-verified research.

EDUCATION

- **Bachelor of Technology in Information Technology (Minor in ECE) – CGPA: 7.72**
  MEA Engineering College | Nov 2021 – May 2025
- **Higher Secondary – 94.58%**
  RMHSS Melattur | 2019 – 2021

PUBLICATIONS

- **A, Shabana.** (2026). **Mitigating Semantic Hallucinations in Zero-Shot Medical Object Detection via Disambiguated Prompt Engineering (1.0).** *Zenodo (CERN Digital Repository).* https://doi.org/10.5281/zenodo.18918582
- **A, Shabana.** (2026). **Real-Time Multi-Class Object Detection and Analytics using the YOLO26n Architecture.** *Zenodo (CERN Digital Repository).* https://doi.org/10.5281/zenodo.18906733
- **A, Shabana.** (2026). **YOLOv11-Powered MRI Classification & Segmentation.** *Zenodo (CERN Digital Repository).* https://doi.org/10.5281/zenodo.18898503

SKILLS

- Programming Languages: Python, SQL.
- Machine Learning: Scikit-learn.
- Deep Learning: TensorFlow, Keras, PyTorch, Ultralytics, Zero-Shot Learning
- Computer Vision: YOLOv11, YOLO26n, Image Segmentation, Object Detection, OpenCV, YOLO-World, Prompt Engineering.
- Data Analysis: Pandas, NumPy, Power BI, Matplotlib.
- Tools: Streamlit Cloud, GitHub, Zenodo, Canva, ORCID.
- Soft Skills: Technical Leadership, Team Collaboration, Adaptability, Problem-Solving.

PROJECTS

Safe Zero-Shot Medical Diagnostics (YOLO-World) | Paper

- Developed the Disambiguated Prompt Engineering (DPE) framework to mitigate "Semantic Hallucinations" in radiological AI.
- Documented and corrected Semantic Alias Bias, allowing zero-shot models to recognize technical instruments via everyday synonyms.
- Implemented anatomical negative prompting to eliminate false-positive structural misidentifications in X-ray scans.

YOLO26n Edge Analytics System | Paper | Live App | GitHub

- Trained an optimized YOLO26n model on a 5,000-image COCO subset via custom CPU pipelines.
- Built a hybrid Streamlit app for real-time local webcam and stable cloud image processing.
- Designed a 2026 "Glassmorphism" UI with live analytics and automated mAP@50 evaluation.

Brain Tumor Diagnostic Suite (YOLOv11) | Paper | Live App | GitHub

- Integrated YOLO11 Classification (**98.7% accuracy**) and Segmentation models for simultaneous diagnosis.
- Implemented a **"Gatekeeper" workflow** to verify classification results before segmentation to prevent false positives.
- Optimized headless environment deployment for real-time probability charts and diagnostic metrics.

MNIST Digit Recognizer | Live App | GitHub

- Developed a custom CNN for handwritten digit recognition with a responsive drawable canvas interface.
- Built a full preprocessing pipeline: grayscale inversion, cropping, and 28×28 normalization.

Language Detection System | Live App | GitHub

- Built a multilingual model using TF-IDF character vectorization, achieving **95.6% accuracy** with Linear SVM.
- Balanced a massive dataset (4,000 sentences/language) to eliminate bias in non-Latin script detection.

EXPERIENCE

Intern – ICFOSS | Sep 2024 – Aug 2025

- Maintained IoT-based rainwater detection hardware; conducted inspections and component replacements.

Freelance Canva Educator | May 2024 – June 2025

- Designed and delivered a self-paced graphic design course; created instructional templates for beginners.

CERTIFICATIONS & LEADERSHIP

- **Python Data Science, AI/ML & Power BI:** Luminar Technolab, Cochin (7-month hands-on training) | 2025 – Present
- **NPTEL:** Principles of Communication Systems (IIT Kanpur) & Digital Circuits (IIT Kharagpur).
- **Leadership:** Department Secretary, Student Placement Representative, and Class Representative. Served as a liaison for faculty-student academic and administrative matters.

LANGUAGES: English, Malayalam, Hindi, Tamil